Haohan (Frank) Fang

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EDUCATION

Columbia University New York, NY
M.S. in Computer Science *Expected Dec 2027*
Focus: Vision, Graphics, Interaction, and Robotics (VGIR)

New York University New York, NY
B.A. in Mathematics and Computer Science (joint major); Minor in Game Design *Sep 2022 - May 2026*

PUBLICATIONS

H. Fang et al. (2026). "Visual-Auditory Proprioception of Soft Finger Shape and Contact." IEEE International Conference on Robotics and Automation (ICRA 2026). [Paper]

RESEARCH EXPERIENCE

Multimodal Audiovisual Perception for Soft Robotic Grippers New York, NY
Research Assistant to Prof. Chen Feng, AI4CE Lab, NYU *Feb 2025 - May 2026*

- Co-developed DeepCoFi, a multimodal proprioception framework fusing RGB imagery and acoustic mel-spectrograms through dual ResNet-18 encoders and a two-stage FoldingNet decoder to reconstruct soft-finger geometry and contact deformation as 2,025-point 3D surfaces.
- Curated 1,617 configurations spanning 20 bending states, 24 contact locations, two indentation depths, and single- and multi-contact cases; achieved 1.51 mm Chamfer error and 47% lower contact error against a DeepSoRo-based baseline.
- Designed an SLA-printed robotic exoskeleton and dual-finger test platform with independent servo actuation and synchronized audiovisual sensing; conducted 600 randomized grasps across 10 objects and achieved 85% test accuracy in downstream object classification.

PROJECTS

Robot Arm Motion Planning and Simulation Sep 2026 - Present
Python, PyBullet, NumPy

- Engineered a 7-DOF robotic manipulation simulation integrating forward kinematics, Jacobian-based inverse kinematics, and collision-aware configuration-space modeling for constrained robot motion.
- Implemented damped least-squares IK and joint-space P/PI feedback controllers to address kinematic singularities, regulate joint motion, and track target trajectories.
- Built a sampling-based motion planning pipeline (PRM, RRT, A* search, randomized shortcutting, B-spline interpolation) achieving a 98% collision-free planning success rate across 100 randomized start-goal queries in scenes with 10-20 obstacles.

Geometric Visual Perception System Sep 2026 - Present
Python, OpenCV, NumPy

- Built a geometry-aware visual perception pipeline integrating image preprocessing, gradient-based feature extraction, and geometric transformations for object localization and multi-view image alignment.
- Implemented gradient-guided Hough voting for line and circular object detection, with sparse accumulators, adaptive thresholding, and non-maximum suppression.
- Developed image registration and mosaicking using homography estimation, RANSAC outlier rejection, inverse perspective warping, and distance-weighted blending for panoramic reconstruction.

INDUSTRY EXPERIENCE

Petch LLC New York, NY
Software Engineering Intern - ML Data Platform *Jun 2025 - Aug 2025*

- Built an end-to-end ML data pipeline that ingested in-app video and sensor streams, segmented recordings into model-ready samples, removed corrupted frames, and exported sharded datasets for downstream training.
- Defined dataset schemas and generated per-sample metadata for automated validation, experiment traceability, and reproducible model training.

SKILLS

Keywords: Multimodal Sensor Fusion, 3D Point-Cloud Reconstruction, Proprioception, Computer Vision, Motion Planning, Robot Kinematics
Languages: Python, Java, JavaScript, C; working knowledge of C++, TypeScript, SQL
ML / Robotics: PyTorch, OpenCV, Open3D, PyBullet, NumPy, Pandas
Tools: Git, Docker, Linux, Flask, React Native, Express.js, Spring Boot, Jenkins
Spoken: Mandarin (native), English (fluent)